

Exploratory Data Analysis – EDA Report

DecodeLabs Industrial Training Kit | Data Analytics – Project 2 | Prepared by Pritesh Raj

1. Problem Statement

This EDA investigates 1,200 e-commerce orders (Jan 2023 – Jun 2025) to answer: **Which products drive the most revenue? What are the key patterns in order value, referral source, and order status? Are there anomalous transactions?** The goal is to transform raw data into actionable business intelligence.

2. Methodology

IPO Framework applied: (a) **Input** – cleaned e-commerce dataset (14 columns). (b) **Process** – descriptive statistics, distribution analysis, trend analysis, outlier detection via IQR method, and Pearson correlation analysis. (c) **Output** – verified insights and business recommendations.

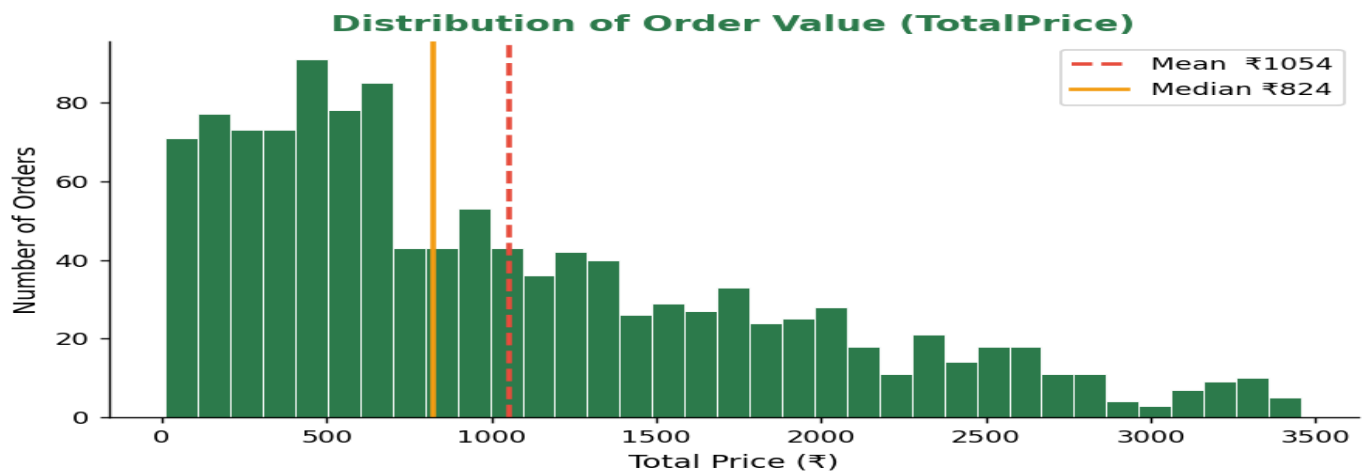
3. Descriptive Statistics

Descriptive Statistics - Numeric Columns

	Quantity	UnitPrice	TotalPrice	ItemsInCart
count	1200.0	1200.0	1200.0	1200.0
mean	2.95	356.41	1053.97	5.48
std	1.41	197.18	819.86	2.28
min	1.0	11.39	11.39	1.0
25%	2.0	186.06	410.52	4.0
50%	3.0	364.21	823.62	5.0
75%	4.0	521.57	1578.48	7.0
max	5.0	699.93	3456.4	10.0

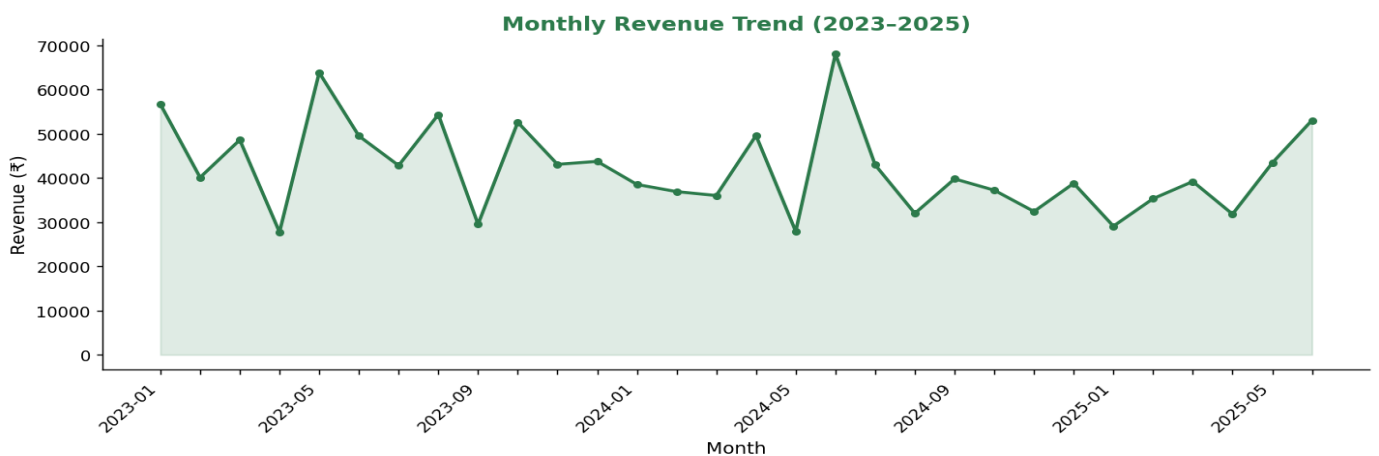
The average order value (mean ■1053.97) is significantly higher than the median (■823.62), confirming a **right-skewed distribution** — a few high-value orders pull the mean upward. For business decisions, the median is a more reliable measure.

4. Order Value Distribution



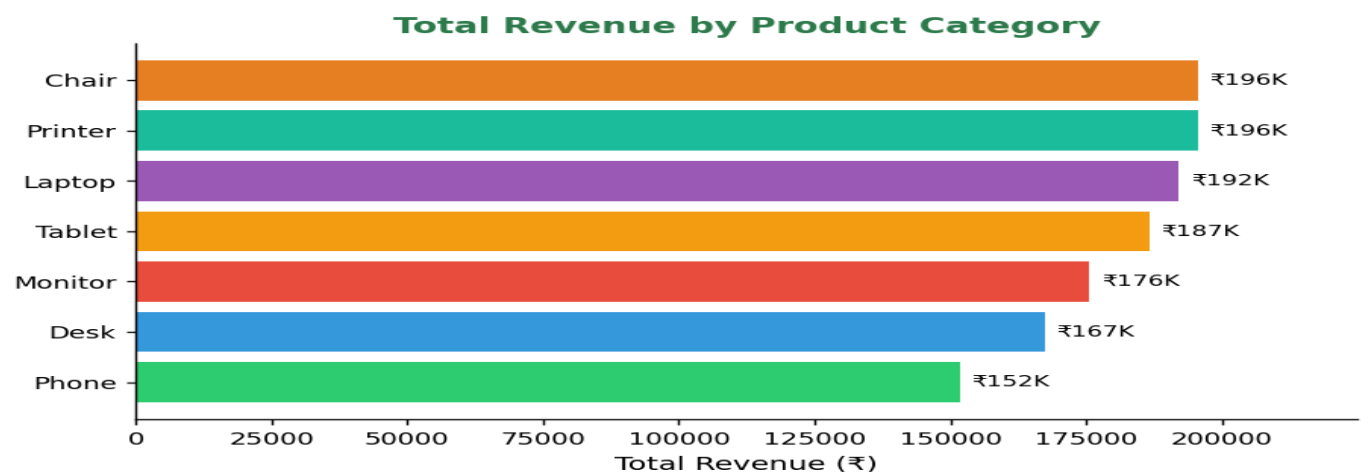
The histogram confirms a right-skewed distribution. Most orders fall between ₹200–₹1,500, with a long tail of high-value orders. This suggests a mix of budget and premium customers.

5. Monthly Revenue Trend



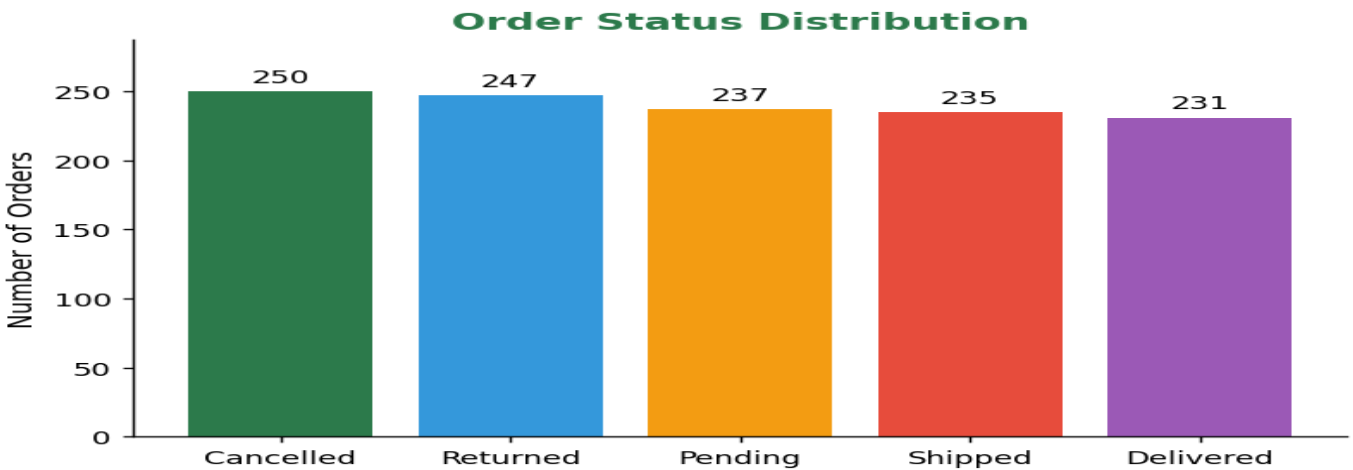
Revenue shows **high monthly volatility** with no consistent seasonal peak. The highest single month was June 2024 (₹68,069). Revenue in 2023 was generally higher than 2024, suggesting a possible decline in order volumes that warrants investigation.

6. Revenue by Product Category



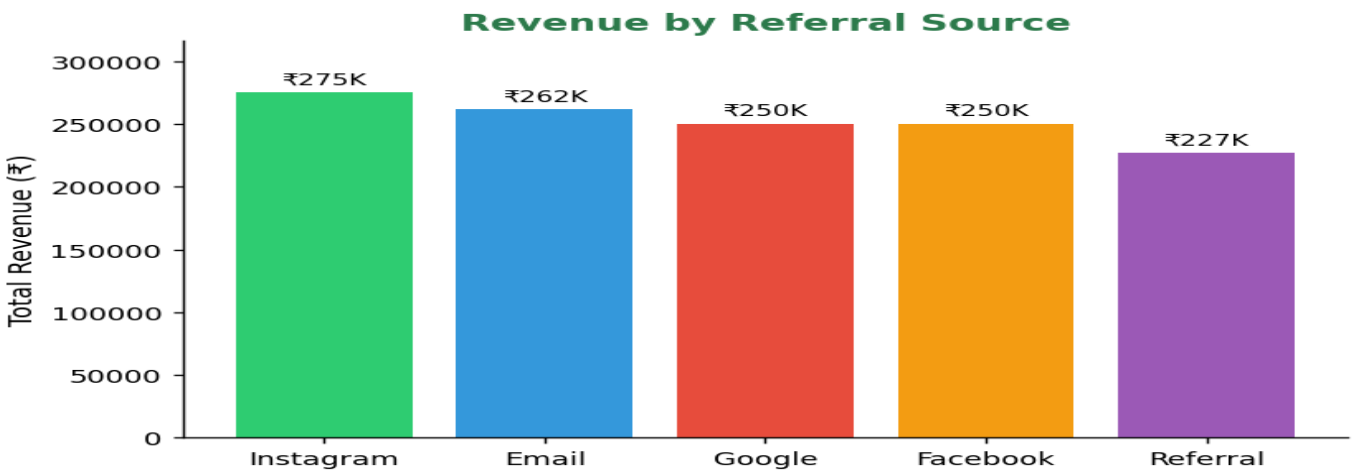
Chair (■196K) and Printer (■196K) are the top revenue-generating products, though all 7 categories are closely competitive. Phone (■152K) is the lowest — possibly due to lower unit prices rather than low demand.

7. Order Status Analysis



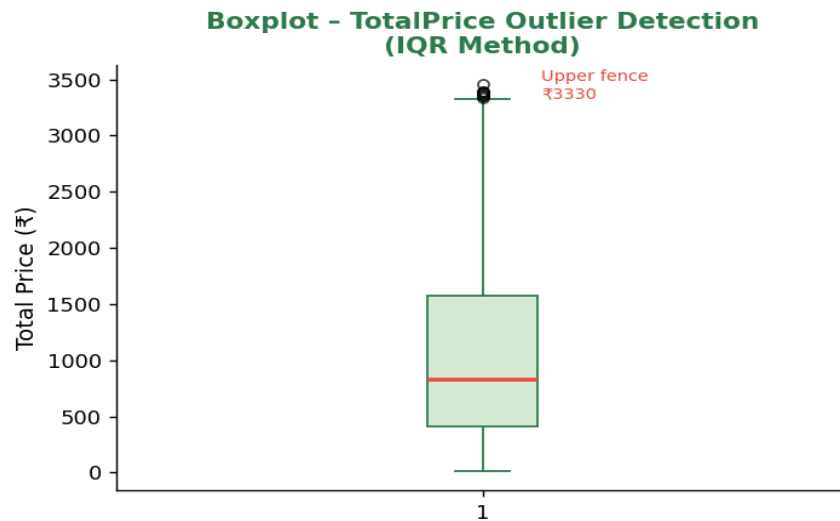
■ **Critical Finding:** Cancelled (20.8%) and Returned (20.6%) orders together account for ~41.4% of all orders. This is an extremely high churn rate and represents significant revenue loss. Business recommendation: investigate root causes of cancellations immediately.

8. Revenue by Referral Source



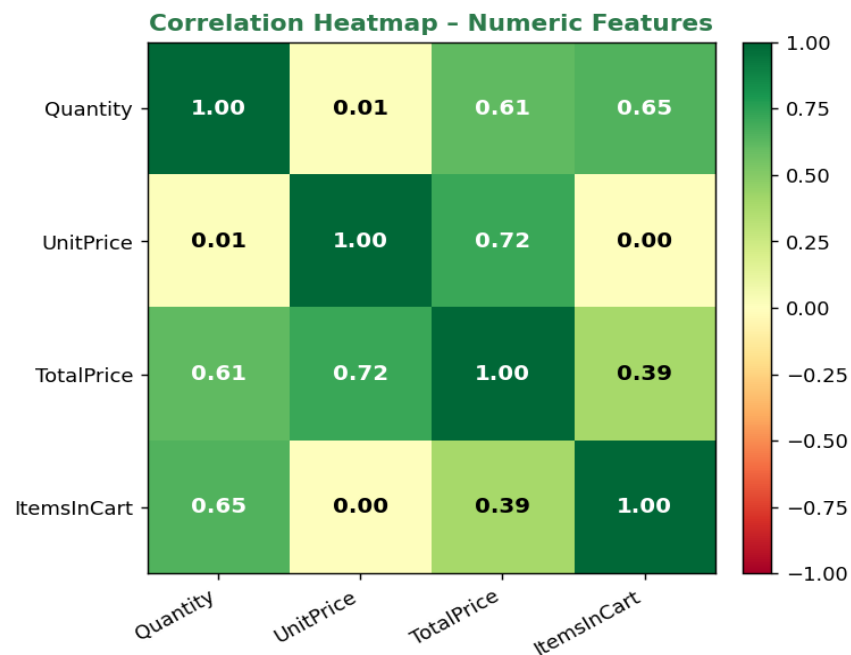
Instagram (■275K) is the highest revenue-driving channel, followed by Email and Google. Referral program generates the least revenue (■227K). Marketing budget should be prioritized toward Instagram and Email campaigns.

9. Outlier Detection (IQR Method)



Using the IQR method ($Q1=₹411$, $Q3=₹1578$, $IQR=₹1168$): Upper fence = $₹3330$. **8 orders were identified as outliers** (values above $₹3,330$). These are likely high-volume bulk orders ($Qty=5 \times \text{high UnitPrice}$) — they are **SIGNAL** (VIP/bulk buyers), not data errors, so they should be retained but investigated separately.

10. Correlation Analysis



Variable Pair	Correlation (r)	Interpretation
UnitPrice ↔ TotalPrice	0.72	Strong positive – higher-priced items drive revenue
Quantity ↔ TotalPrice	0.62	Moderate positive – more items = higher bill
ItemsInCart ↔ Quantity	0.65	Moderate – browsing behaviour linked to purchase qty
UnitPrice ↔ Quantity	0.01	No correlation – price doesn't affect quantity ordered

Note: Correlation ≠ Causation. These are clues, not verdicts.

11. Key Findings & Business Recommendations

#	Finding	Recommendation
1	Chair & Printer lead revenue (■195K each)	Increase inventory & run targeted promotions
2	Cancelled+Returned = 41.4% of orders	Urgent investigation — implement post-purchase surveys
3	Instagram is the #1 revenue channel (■275K)	Increase marketing budget allocation to Instagram
4	Mean (■1054) >> Median (■824) — right skew	Use median for pricing strategy; target mid-market segment
5	8 bulk/VIP orders above ■3,330 detected	Create a VIP customer segment with loyalty benefits
6	UnitPrice drives TotalPrice most strongly (r=0.72)	Premium product upselling will have highest revenue impact

Tools used: Python 3, Pandas, NumPy, Matplotlib, ReportLab